

Architectural Particular Specification
SIGNAGE
SECTION 10.1

Rev	Date	Description
0	14 APR 2022	First Issue

SECTION 10.1

SIGNAGE

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The work of this section includes, but is not limited to, the furnishing and installation of the following:
- B. Fabrication of signage as indicated on the Drawings
- C. Requisite professional transportation and protection from site of fabrication to site of installation
- D. Installation services including permanently fixing all signage. This includes, but is not limited to, the following:
 - 1. Exterior signage
 - 2. All interior room tags or graphic letters
 - 3. All interior indication graphic on doors (such as lavatory doors)
 - 4. All interior directional graphics
 - 5. All interior wall mounted lift lobby identification and service messages
 - 6. All statutory signs
- E. Closely align and verify all dimensions, tolerances, clearances, electrical load requirements and other particulars such that the signage fits seamlessly with other works and all adjacent finishes.
- F. Provide final connection from the fused spur to the signage in accordance with the EMSD and FSD Code of Practice and Regulations if lightings are required.
- G. Provide Structural Engineering design, calculation and shop drawings with registered structural engineer endorsement if required.

1.02 REFERENCES

- A. BSI – British Standards Institute:
 - 1. BS 219: 1977 (1989)-Specification for Soft Solders.
 - 2. BS639 - Specification for Covered Carbon and Carbon Manganese Steel Electrodes for Manual Metal Arc Welding.
 - 3. BS 729: 1986 - Specification for Hot dip Galvanised Coatings on Iron and Steel Articles.
 - 4. BS 1449 (a): Part 1 – Steel Plate, Sheet and Strip.
Part 2 – Specification for Stainless and Heat Resisting Steel plate, Sheet and Strip.
 - 5. BS 1453: 1987 - Specification for Filler Materials for Gas Welding.
 - 6. BS 1470: 1987 - Specification for Wrought Aluminium and Aluminium Alloys for General Engineering Purposes - Plate, Sheet and Strip.
 - 7. BS 1471: 1972 - Specification for Wrought Aluminium and Aluminium Alloys for General Engineering Purposes - Drawn Tube.

8. BS1473: 1972 - Specification for Wrought Aluminium and Aluminium Alloys for General Engineering Purposes - Rivets, Bolt and Screw Stock.
 9. BS 1474: 1987 - Specification for Wrought Aluminium and Aluminium Alloys for General Engineering Purposes : Bars, Extruded Round Tubes and Sections.
 10. BS2910: Part 2: 1990 - Filler Rods and Wires for Gas-shielded Arc Welding: Stainless Steel.
 11. BS3111: Part 2: 1979 - Specification for Cold Forged Fasteners and Similar Components - Stainless Steel.
 12. BS4127: Part 2 : 1972 (1986) - Specification for Light Gauge Stainless Steel Tubes (metric unit).
 13. BS4360: 1990 - Specification for Weldable Structural Steels - Plates, Strip, Flats and Bars.
 14. BS5950 - Structural Use of Steelwork in Building.
 - a. Part 2: Specification for Materials, Fabrication and Erection: Hot Rolled Sections.
 - b. Part 7: Specification for Materials, Fabrication and Erection: Cold formed Sections.
 15. BS7364: Part 1: 1991 - Cathodic Protection. Code of Practice for Land and Marine Applications.
 16. BS7079 - Preparation of Steel Substrates before Application of Paints and Related Products.
- B. BS EN – European Standard:
1. Occupational and Fire Safety Standards-BS EN 341, BS EN 344, BS EN 353, BS EN 361, BS EN 363, BS EN 367 and other applicable codes and standards.
- C. American Standards:
1. AAMA No. 605.2 -90 - Voluntary Specification For High Performance Organic Coatings on Architectural Extrusions and Panels.
- D. Except as modified by governing codes and by the Contract Documents, comply with the applicable provisions and recommendations of the following:
1. AWS D1.1 "Structural Welding Code - Steel".
 2. AWS D1.3 "Structural Welding Code - Sheet Steel".
 3. NAAMM "Metal Finishes Manual".
 4. Industrial Fasteners Institute "Fastener Standards Book."
 5. AAMA 2603 "Voluntary Specification, Performance Requirements and Test Procedures for Pigmented Organic Coatings on Extruded Aluminum and Panels".
 6. PD 6484: 1979(1990)-Commentary on Corrosion at Bimetallic Contacts and Its Alleviation.
- E. Local Codes:
1. Design Manual - Barrier Free Access 2008
 2. Code of Practice on Wind Effects-Hong Kong, 2004
 3. Practice Notes (PNAP)
 4. FSD code of Practice
 5. Building Regulations

6. Building Ordinance

1.03 QUALITY ASSURANCE

- A. Propose sub-contractor specializing in signage fabrication and installation with minimum 5 years continuous experience.
- B. The specialist sub-contractor should be able to prepare shop drawings and samples, fabricate and install the works shown in the Drawings.
- C. The specialist sub-contractor should demonstrate ability with a complete project record satisfactory to the Designer.

1.04 PERFORMANCE REQUIREMENTS.

- A. Works shall conform with all relevant statutory ordinances, regulations, orders, codes of practice in force at the time of signing of the Contract.
- B. The Contractor shall allow for site tolerances, discrepancies and changes in the building layout, where the principles and intent of the signs are not affected.
- C. Weather resistance
 - 1. Works must be fully capable of resisting atmospheric conditions prevailing at the site, which, may be uncontrolled and in some areas the temperature and/or relative humidity may exceed external conditions. In particular no condensation shall take place either on the internal surfaces or within the interstitial construction thickness of materials used in the signs;
 - 2. Externally located powered signs or powered signs not in environmentally controlled conditions shall be suitable for exposure to wind loading, ultraviolet and ozone exposure, high humidity and temperature, vandalism, and pest infestation;
 - 3. Internally located signs shall be designed to withstand the temperature and humidity conditions of the interior environment.
- D. Assembly
 - 1. The Contractor shall endeavour to produce the signs with a minimum number of variable components and ensure that the signs may be installed using the simplest possible methods feasible;
 - 2. All similar items of equipment and their component parts shall be completely interchangeable. Spare parts shall be manufactured from materials similar to the originals and shall fit all similar items or plant. Where machining may be needed before fitting renewable parts, the machining fits together with their tolerances shall be shown on the shop drawings.
 - 3. Fixings shall not be visible, unless otherwise specified in the Drawings. Where fixings are visible, their appearance shall match or suit the items being fixed and the elements to which they are fixed. The method of fixing shall not damage anything being fixed or anything receiving fixings. Equally, items which require accessibility or removal shall have fixings that facilitate such accessibility or removal. All exposed screws, fastenings shall be of appropriate materials to the approval of the Architect.

E. Accessible parts

1. The Contractor shall design all moveable parts and mechanisms (such as removable or hinged panels) to be maintenance-free wherever possible, or be located at positions where maintenance to these parts can be reached without dismantling the signs support or structure;
2. All illuminated signs shall have access panels and casings allowing maintenance of lamps and parts;
3. All signs shall allow access to internal electrical components yet be lockable so as not to be accessible to the general public;
4. All mechanisms shall be fail-safe and demonstrated as such to the Architect by testing a prototype;
5. Safety restraints to all movable assemblies shall be built-in to prevent accidental dislodgement and subsequent collapse during maintenance.

F. Powered signs

1. Powered signs shall have their on/off control linked to the control of the building management system. Power supply shall be non-essential type except for signs required as fire services installation which shall be supplied by essential power supply;
2. The Contractor shall be responsible for the supply and installation of suitable light fittings to illuminated signs including concealed type wiring connections to power supply points provided by the Main Contractor, and for detailing of lighting fittings to obtain a satisfactory and even illumination effect;
3. All electrical and wiring work shall comply with IEE Regulations and the Power Company's supply rules and regulations. Internal illumination shall be provided by low energy fluorescent lamps simply fitted into the signs unless otherwise specified. Adequate ventilation shall be provided in accordance with lamp manufacturer's recommendations while there shall be no light leakage;
4. All illuminated signs which are to be installed in an exposed or semi-exposed location shall be sealed to a rating of IP55 to BS EN 60529 Degrees of protection provided by enclosure or equivalent ASTM standard and be provided with at least 2 holes placed so as to allow condensation to drain away without significant light leakage. A fine stainless steel insect filter shall be fitted to prevent infestation;
5. Quantity and positions of fluorescent tubes must be sufficient and effective enough to illuminate all wordings and graphics clearly;
6. The Contractor is responsible for all electrical wiring from the nearest switch / plant rooms to the lighting, electrical gear, power units, etc. required for the relevant signs, with the exception of signs for individual shops. An individual fuse link shall be provided inside each individual illuminated sign. The Contractor is responsible for the safe connection of wiring to signage, and the concealment of wiring. The Contractor is required to submit the relevant electrical wiring and conduit shop drawings to the E/M consultants for approval prior to commencement of any fabrication works.

G. Loading capacity

Design, fabricate, and install each of the sign to withstand all normal deflections and tolerances of the structure to which it is fixed, under all design loads or combination of loads, without damage or any reduction in its performance. Design shall allow for sufficient design, manufacture, and installation tolerance of all interfacing conditions, to include allowing for any movement arising from differential superimposed loads, dead loads, self-weight, variations in moisture conditions, thermal expansion and contraction, deflection from external forces and torsional loads, deflection from excessive temporary loads during construction and stacking of materials and all other

relevant forces. Signs shall be capable of accommodating their own dead load. The Contractor shall confirm that no fracture, dislodgement, measurable nor visible deformation nor any deterioration of the subsequent performance of any part of the signs will result from such part being subjected to typical design loads. Signs shall withstand all reasonable static and dynamic loads placed upon them without causing permanent deformation of components or the failure of members or opening of joints and shall transmit such loads safely to the points of support.

H. Bonding

The Contractor shall ensure that all extraneous conductive parts of the signs are effectively bonded to earth. Each component shall constitute an extraneous conductive part.

I. Impact resistance and wearing

1. All painting works over any metal surface shall have adequate hardness/rigidity to resist the impacts from any hand-held luggage / objects without any noticeable damage;
2. All signs and graphics should be able to withstand wearing, without any noticeable damage, resulted from the daily normal maintenance works of the areas the signs locate.

J. Statutory Signage

1. All signs required under the statutory regulations and guidelines, e.g. exit signs and fire hose reel signs, should comply with all relevant statutory requirements;
2. Where notices / signages are required on Drawings / Specifications for indicating the Fire Service Access Point (such as access to fireman's lift), they should be of a type of safety sign which is self-energised / self-powered in respect of luminosity and which require no external source of power. The products shall also comply with BS 5499: Part 2 and to the satisfaction of the relevant authorities including Fire Services Department, Buildings Department.

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTAL

- A. Product Data: manufacturer's technical data, copy of information on paint label, and instructions for handling, storage, installation, protection, and maintenance of each product.
- B. Submit Material Safety Data Sheets (MSDS Sheets) indicating health risks, flammability, handling and storage precautions for items required under this Section.
- C. Samples: Submit samples of each material, finish, colour, accessories and lettering type required prior to the construction of any full size prototypes:
 1. Typical typography spacing template for signs.
 - a. Natural Finishes: two 300mm x 300mm samples of each specified metal, stone and glass specified demonstrating typical folding and welded and non-welded joint details as specified in the Tender Drawings.
 - b. Paint: two 300mm x 300mm samples of each specified paint colour, on the actual substrate to be used.

2. Hardware/ Fasteners: two samples of each visible hardware element and fastener.
 3. Print-outs showing portion of actual size directory graphic.
- D. Full Size Prototypes:
1. Prior to approval of final shop drawings supply one full size, non returnable and fully operational sample of the different sign types. And all samples to be reviewed at factory prior to Client submittal on site.
- E. Typography: One paper pattern of each directional sign per A3 sheet.
1. Show exact typeface, letter spacing, symbols, arrows, and positioning.
 2. Show solid black letterforms and graphic elements on a white background. Outlines will not be accepted.
 3. Outline the sign face.
- F. Shop Drawings:
1. For signs that require engineering, submit structural calculations and shop drawings produced and stamped by an HKSAR Registered Structural Engineer with knowledge of the relevant Hong Kong authorities regulations.
 2. Show details of fabrication, including materials, dimensions, finishes and methods of construction.
 3. Show details of installation, including materials, dimensions, finishes, anchorages, and methods of construction.
 - a. Where installation method is not specified in detail, show proposed method of installation.
 - b. Furnish location template drawings for supports or anchors to permanent construction.
 4. Specify fasteners and detail methods of fastening.
 5. Specify lamp dimensions and manufacturers. Provide technical data for control gear.
 6. Electrical wiring drawings.

1.08 DELIVERY, STORAGE, AND HANDLING

- A. Protect, store and handle signage in manner so as to prevent damage and marking on exposed surfaces and finishes.
- B. Package separately or in like groups of names, labelled as to names enclosed; include installation template, attachment system and installation instructions. Uniquely label each packaged sign for distribution on site.
- C. All signage components to be securely packed and protected within wooden crates for protection during shipment, storage, and installation.
- D. Protect signs and all adjacent surfaces and materials from damage during installation.
- E. Signage Subcontractor shall agree with the Management Contractor on a lockable delivery/ staging area for signs and installation materials.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

(NOT USED)

1.11 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS

(NOT USED)

2.02 MATERIALS

A. General Requirement

1. Provide materials of colours and textures specified herein and indicated on drawings.
2. Like materials shall have same finishes, uniform surfaces, visually uninterrupted by sections, hardware, or mounting methods, unless otherwise indicated.
3. Colours: Consistent throughout entire length and edges of signs, as well as throughout the entire Subcontract.
4. Tolerance Criteria: Surfaces shall be flat to within plus or minus 0.75mm when measured at any point with 1metre straight edge.
5. All materials selected shall be suitable for purpose and satisfies the requirements for HKSAR Codes such as fire integrity. The design of housings for internally illuminated signs shall take into consideration provision of adequate ventilation to prevent heat build up within.

B. Metals

1. Aluminium and Stainless steel, sheet:
 - a. Panel thickness: as specified in drawings. Where not specified, use thickness sufficient to remain rigid in use without deformation or "oil canning".
 - b. Surfaces shall be true, smooth, and free of extrusion marks or blemishes.
 - c. Finish all surfaces to protect against corrosion.
 - d. All folded surfaces require a v-notch detail on non-visible surfaces prior to fabrication.
 - e. Cut out lettering, symbols and shapes must not deform adjacent surfaces.
2. Metal for Framing and Bracing: shape, dimensions, and alloy temper necessary to meet engineering requirements and the relevant Hong Kong authorities building codes.
3. Finishes for Aluminium and Stainless Steel:
 - a. Painted: Surface primed as necessary for paint adhesion.
 - b. Hairline: Hairline finishes matching Designer's specification.
 - c. Polished: Mirror polished.
4. Tubing: seamless.

C. Glass:

1. Clear laminated glass must be of an international standard.

D. Screen print Ink:

1. Colours: Match specified colours; no colour variations or substitutions will be accepted.
2. Finish: matte.

E. Inkjet Outputs:

1. All inkjet outputs should be high resolution to match photographic quality.
- F. Powder Coat Process and Paints:
1. All applied finishes shall be specified and applied by the Signage Subcontractor. These finishes must not fade, chalk, crack or deteriorate.
 2. Paint: linear polyurethane, chemically cured, two-component paint.
 3. Primers: as recommended by paint manufacturer.
 4. Clear Coat:
 - a. Apply a final clear coat over pigment coat.
 - b. Finish: Unless otherwise specified, eggshell (matte) finish having an 11 to 19 degree gloss on a 60-degree glossimeter.
 5. Colours: Match specified colours; no colour variations or substitutions will be accepted.
- G. Plastics: Acrylic
1. Acceptable Manufacturer: Perspex, American Cyanamid, or approved equal.
 2. Use solid sheet, laminated sheets, or cast acrylic, in size, thickness, and opacity specified on sign drawings or in sign schedule.
 3. Finish: exposed surfaces shall be eggshell (matte) finish.
- H. Plastics: Adhesive-Backed Vinyl Sheeting for Cut Graphics:
1. All specified vinyl materials are available from:
3M Hong Kong Limited
5/F Victoria Centre
15 Watson Road, Hong Kong
Telephone: (852) 2806 6256 (Direct)
Facsimile: (852) 2807 0976 (Direct)
Or equivalent
- I. Panaflex:
1. All specified Panaflex materials are available from:
3M Hong Kong Limited
5/F Victoria Centre
15 Watson Road, Hong Kong
Telephone: (852) 2806 6256 (Direct)
Facsimile: (852) 2807 0976 (Direct)
 2. Or equivalent
- J. Adhesives and Tapes
1. Vinyl adhesive: low-tack, removable.
 2. Foam Tape:
 - a. High-density, open-cell, polyurethane double-faced.
 - b. Manufacturer: 3M, or approved equal.
 - c. Colour: as specified.
 3. Isotac Tape: Isotac P.S. ADH. 10 Mil by Scotch 3M, #Y-9473.
 4. Plastic Adhesives: type specified by the plastic manufacturer.

- K. Fasteners:
 - 1. Cadmium-plated, except stainless steel.
 - 2. Tamperproof in exposed locations, unless otherwise noted.
 - 3. Stainless steel for assembling ferrous to non-ferrous metals.
 - 4. Exposed fasteners: match colour and finish of adjacent sign surfaces unless otherwise noted.
- L. Hardware
 - 1. Hinges:
 - a. Stainless steel.
 - b. Exposed surfaces: match colour and finish of adjacent sign surfaces unless otherwise noted.
- M. Locks:
 - 1. Rust-proof.
 - 2. Internal mechanism: top-quality pin tumbler type.
 - 3. Exposed surfaces: match colour and finish of adjacent sign surfaces unless otherwise noted.
- N. Electrical Components
 - 1. All lamps and control gear must be of an international standard, i.e. Osram, Philips, GE Lighting, etc. All sources must be supported by technical data.
 - 2. All lamps to be supplied from a single source where practicable.
 - 3. Waterproof fittings must be used for signs in outdoor environments.
 - 4. Signage Subcontractor to incorporate ventilation provisions as necessary for proper functioning of the signage without any light leakage.
- O. Adhesives and Accessories
 - 1. Provide as indicated, as listed on signage schedule, and as required for complete signage installation; non-toxic adhesives.

2.03 FABRICATION

- A. General:
 - 1. Fabricate to conform with the drawings and control samples; finish exposed surfaces unless otherwise indicated.
 - a. Verify and be responsible for all dimensions and conditions shown by sign drawings. Field measure before shop drawing preparation and fabrication. Written dimensions on drawings take precedence over scaled dimensions.
 - b. Notify Designer of any discrepancies in the drawings, sign schedule, field dimensions or conditions, and of changes required in construction details.
 - c. Provide and integrate all materials and parts necessary to complete and install each sign even though such items or work are not specifically shown or specified.
 - d. Signs in outdoor locations to be provided with drainage provision.
- B. Workmanship:
 - 1. General:

- a. Isolate ferrous from non-ferrous metals to prevent electrolysis. Fasten with stainless steel fasteners.
 - b. Design and fabricate signs as required to accommodate uneven rates of thermal expansion and other movement.
 - c. Structural and functional elements of internally illuminated signs shall not cast shadows on externally visible illuminated areas.
 - d. Follow manufacturers' recommendations for fabrication, fastening, integrating, and finishing of all materials, hardware, electrical and component parts.
- 2. Quality:
 - a. Execute and finish Work in accordance with approved drawings, cuts, details, and samples.
 - b. Form signs to shape and size with sharp lines, angles, and true curves. All folded metal surfaces require a v-notch detail on non-visible surfaces prior to fabrication.
- 3. Finished Surfaces:
 - a. Exposed surfaces: smooth, blemish-free surfaces unless otherwise specified.
 - b. Edges: eased and free of burrs.
 - c. Surfaces intended to be flat should be free of bulges, oil canning, or other deformities.
 - d. Finish welds on exposed surfaces shall be imperceptible in the finished work.
- 4. Fasteners:
 - a. Fully conceal fasteners, hinges, and brackets, except where shown on drawings or otherwise approved by Designer.
 - b. Holes for visible fasteners must have clean edges.
- 5. Fit:
 - a. Where a tight fit is required, make joints close.
 - b. Fabricate access panels tight-fitting, weatherproof, and flush with adjacent surfaces.
 - c. There shall be no light leaks from internally illuminated signs.
- 6. Labelling:
 - a. Conceal all UL and identification labels inside signs.
 - b. Do not apply labels which cannot be concealed, except those required by regulations in locations approved by Designer.
- C. Vinyl Letters and Graphics: Kiss cut using computer generated copy.
- D. Silkscreen Letters and Graphics: Computer generate screens for signs and graphics to designs and criteria established during sample submissions.
 - 1. Artwork: Prepare artwork based upon Designer's layout.
 - 2. No substitutions of typefaces, symbols, or art will be accepted.
- E. Shop Fabrication: Shop fabricate signs for Designer review prior to packing; disassemble signs only as required for shipping and handling.
 - 1. Clean signage prior to packing and shipping.

F. Typefaces, Symbols and Artwork

1. General:

- a. Use typefaces, symbols and artwork specified. No substitutions will be accepted without prior approval by Designer.
- b. Reproduce typography, symbols, and arrow by photo process. Hand-drawn reproductions are not acceptable.
- c. Ensure correct sizing and positioning of typography, based on sign drawings and specifications. Place typography, symbols, and arrows on signage in the size/ position relationships shown on drawings.
- d. Ensure sign graphics are sharp, clean, and free of ticks.

2. Typography:

- a. Use information on Designer furnished disk to insure proper typefaces, letter spacing, and leading on finished signs.
- b. Align letterforms to maintain a base line parallel to the sign format. Extend bottoms of round-bottom letters and of letters with descenders below this baseline according to the standards of the typeface.

3. Symbols:

- a. Provide Designer's symbols as specified.

4. Arrows:

- a. Use only custom arrows as shown on drawings.

G. Metalwork

1. Finish per schedule and approved samples.
2. Welded Seams and Joints: grind, fill, and finish flush and smooth with adjacent surfaces. Welded seams and spot-welded joints shall be not be visible on exterior of signs after final finish is applied.
3. Exposed Seams: where allowed, butt factory-cut edges to produce hairline joints. Exposed seams are not required to be filled.
4. Cut out lettering, symbols and shapes must not deform adjacent surfaces.
5. All folded metal surfaces require a v-notch detail on non-visible surfaces prior to fabrication.

H. Plastics / Vinyls

1. Acrylic:

- a. Cut and machine in accordance with manufacturer's recommendations.
- b. Finish exposed edges smooth.

2. Cut Acrylic Letters:

- a. Cut in a manner such that edges and corners of cut letters are true and clean.
- b. Ease edges; remove burrs.
- c. Letterforms with rounded positive or negative corners, nicked or ragged edges, etc., are not acceptable.

3. Panaflex:

- a. Cut and machine in accordance with manufacturer's recommendations.
- b. Seams to be indicated on shop drawings..

4. Vinyl:

- a. Use largest width material at all times, with no seams in finished work.
 - b. Finish: as specified.
- 5. Die Cut Vinyl:
 - a. Cut on brand equipment.
 - b. Artwork: electronically digitised.
 - c. Pre-space messages, symbols, and arrows on application tape, according to approved patterns prior to installation on sign.
- I. Lamps
 - 1. Fluorescent and neon lamp placement must provide an even level of illumination. Light diffusion films to be applied as necessary.

2.04 FINISHES

- A. Paint Application:
 - 1. PVF2 coating /Powder coat /spray apply, where specified.
 - 2. Final colour coat: 3 to 5mm thick.
 - 3. Finish: uniformly coated, free of brush marks, streaks, runs, or pile-ups of paint.
- B. Screen Printing:
 - 1. Screens:
 - a. Use only photo-process screens.
 - b. Materials: nylon or polyester.
 - c. Minimum 250 lines per inch.
 - 2. Use weather-resistant materials compatible with the intended substrates.
 - 3. Ensure clean, sharp, solid edges of screen-printed graphics without edge build-up or bleeding.
 - 4. Finish Surfaces: uniform in colour and finish; free of pinholes or blemishes.
- C. Colours for backgrounds and messages:
 - 1. Colours: as specified. No colour variations or substitutions will be accepted.
 - 2. Match samples approved by Designer.
 - 3. Maintain proper chroma value and coverage.
 - 4. Maintain proper opacity.
 - 5. Free of blistering, bleeding or other imperfections.
- D. Colour registration: crisp, sharp, and free of ticks, line waver, or other imperfections.

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. Examine areas and conditions under which signs are to be installed. Include for all bridging structures, hangers, bolts, etc. to avoid M&E services.
- B. Beginning of installation indicates acceptance of conditions.
- C. Arrange a meeting with Architect and Designer after award of contract.
 - 1. Review locations, heights and alignments of signs
 - 2. Review installation procedures.
 - 3. Review Power requirements and locations.
 - 4. Inform Designer of any discrepancies.
- D. Arrange an on-site meeting with Designer and Client to review completed installation. Examine sign locations, quantities, alignments, and quality of work. Note deviations from drawings or specifications, or work of unacceptable quality, for correction by Contractor.

3.02 INSTALLATION

- A. Securely fasten signage in place at agreed locations and heights using methods shown on drawings and submittals. Use concealed fasteners or adhesives, unless otherwise specified.
- B. General: Install signs free from distortions and defects.
 - 1. Install signage plumb, level, in correct alignment with adjacent construction, as indicated and in exact location as shown on drawings.
 - 2. Install signs after substrates are finished.
 - 3. Securely attach signage to supporting structure, ceiling or walls as indicated with concealed fasteners.
- C. Install signs properly aligned to established lines and elevations, plumb, level and true to line and dimension, clean, and flush to surface unless otherwise specified.
- D. Install in a neat and proper manner. Protect adjacent surfaces from damage.
- E. Make final electrical connections.
- F. Dimensional Letters Signs: Locate letters and spacing in field with full size computer generated installation drawings; Designer shall review final location prior to installation.
- G. Wall Mounted Signs:
 - 1. Install signs on walls and columns after surfaces on which they are to be mounted are finished. Mount signs at heights indicated on drawings. Provide paper templates to Designer, as requested.
 - 2. Templates of signs to be mounted on finished surfaces to be checked on-site by the Architect/ Designer to review final positions for drilling of holes.
- H. Ceiling mounted: Co-ordinate installation with interior fit out contractor to ensure anchor bolts /support rods can be fixed prior to installation of any permanent ceiling finish.
- I. Toilet Room Signs: Install signs on doors after door finishing is completed.

- J. Entry Signs: Install in locations reviewed by Designer.
- K. No on-site polishing, finishing or modification of fabricated signs will be accepted.

3.03 CLEANING

- A. Clean and polish. Remove excess adhesive.
- B. Remove sign installation waste and debris from the site.

3.04 PROTECTION

- A. After installation clean soiled surfaces with maintenance materials using extreme care to prevent damage to signage and to adjacent building surfaces.
- B. Protect signage from damage until Project completion; repair or replace signage damaged by construction activities.
 - 1. Repairs shall match original signage; where the Architect/ Designer considers the rectification work unsuitable, replace signage.
- C. Adjust hardware to assure secure and safe mounting.
- D. Remove protective coatings and identifying labels from signs.
- E. Repair damage incurred during installation to the satisfaction of the Designer and Client.
- F. Replace signs which cannot be repaired to satisfactory condition at Signage Subcontractor's expense.

3.05 MAINTENANCE

- A. Provide Client with 2 complete hard copy and soft copy sets of as-built drawings on completion of project.
- B. Provide Client with written instructions on proper care and maintenance for all signs installed.

END OF SECTION